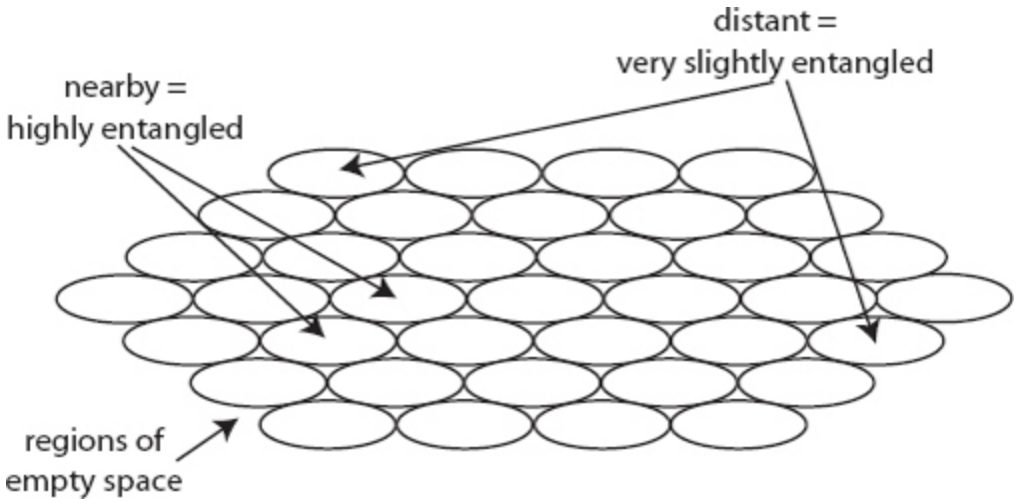


Quantum-mechanically, the space of all the possible wave functions for a system is that system's Hilbert space. So the Hilbert space describing any region in quantum field theory is infinite-dimensional, because there are an infinite number of degrees of freedom. As we'll see, that might not continue to hold true in the correct theory of reality; there are reasons to think that quantum gravity features only a finite number of degrees of freedom in a region. But quantum field theory, without gravity, allows for infinite possibilities in any tiny box.

Those degrees of freedom share a lot of entanglement with the degrees of freedom elsewhere in space. To drive home just how much, imagine starting with the vacuum state, taking one of those one-cubic-centimeter boxes, and poking the quantum fields inside. By "poking" we mean any way we could conceivably imagine affecting the field just in that local region, by measuring it or otherwise interacting with it. We know that measuring a quantum state changes it into another state (indeed, to different states on each branch of the new wave function). Do you think that by poking the state strictly inside a given box, it's possible to instantly change the state outside the box?

If you know a little relativity, you might be tempted to answer "no"—it should take time for any effects to propagate to faraway regions. But then you remember the EPR thought experiment, where Alice's measurement on a spin can affect the quantum state of Bob's spin, no matter how far away they are from each other. Entanglement is the secret ingredient. And we just said that the vacuum state in quantum field theory is highly entangled, such that every box is entangled with every other box. Gradually you will begin